The Complete Structural Mereology of a Neurosymbolic Search-and-Rescue Manifold: A Unified Axiomatic System Specification

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1 Introduction and Architectural Core

We formalize the search-and-rescue plot for a central Fire-Station as an interlocking multi-graph Sequential Dynamic System (SDS) operating across distinct modal boundaries. Following Edward Zalta's Theory of Abstract Objects, the spatiotemporal execution plane maps to ordinary object exemplification (Fx), while the invariant control forests encode properties atemporally (xF). The ultimate objective of the system is the minimization of coordinate uncertainty to realize Mesarovic's Attainable Space (V'), culminating in physical vehicle extraction.

2 The 30-Dimensional Integrated Attribute Matrix

To inject continuous meaning and speech-act context into the message-passing convolutions, the GNN ingests an integrated, multi-scale node attribute matrix $\boldsymbol{X} \in \mathbb{R}^{N \times 30}$:

$$\boldsymbol{x}_i = \left[\boldsymbol{e}_i^{\text{emp (16)}} \mid \boldsymbol{g}_i^{\text{spatial (2)}} \mid \boldsymbol{p}_i^{\text{ppl (3)}} \mid \boldsymbol{r}_i^{\text{rsa (3)}} \mid \boldsymbol{s}_i^{\text{searle (3)}} \mid \boldsymbol{t}_i^{\text{deontic (3)}} \right] \in \mathbb{R}^{30} \tag{1}$$

3 High-Fidelity Electromagnetic Transient (EMT) Systems and IEEE 39-Bus Testing

To ground the continuous-time physics layer into a verifiable industrial framework, sub-millisecond electromagnetic transient (EMT) numerical simulations are executed across a modified IEEE 39-bus test system via the open-source Julia platform `PowerSimulationsDynamics.jl`. Let the transient state of an electrical node i be characterized by a continuous differential-algebraic power vector $\boldsymbol{\xi}_i(t)$:

$$\boldsymbol{\xi}_i(t) = [V_i(t),\, \delta_i(t),\, P_{\text{IBR},i}(t),\, Q_{\text{IBR},i}(t)] \in \mathbb{R}^4 \tag{2}$$

where $V_i(t)$ represents the instantaneous voltage magnitude, $\delta_i(t)$ defines the phase angle deviation, and $P_{\text{IBR},i}(t), Q_{\text{IBR},i}(t)$ monitor the active and reactive power outputs of local inverter-based renewable resources. The sub-millisecond voltage variations are processed natively by the inductive learners as extensional grounding exemplars (`exs.pl`):

$$\mathcal{E}_i \implies \text{Ingest}\left(\boldsymbol{\xi}_i(t_{\text{sample}}),\, \Delta t = 0.5 \text{ ms}\right) \tag{3}$$

This allows our distributed MARL controllers to intercept microscopic transient voltage droops ($V_i(t) < 0.95$ p.u.) and enforce the Sherman-Gable (1983) backoff boundary constraints before sub-synchronous oscillations propagate into catastrophic inter-area blackouts.

4 Symmetric Throughput Invariants and Parameter Estimation

Following the network routing paradigms of **Sherman & Gable (1983)**, our cyber-physical framework evaluates carrier capacity limits across the shared transmission line using a normalized propagation delay parameter $a = \tau_{\text{prop}}/T_{\text{trans}}$. Given electricity propagation speed in copper ($v \approx 2 \times 10^8$ m/s), a campus cable runway length of $L = 1000$ meters results in an end-to-end propagation time of $\tau_{\text{prop}} = 5$ μs. Bound against our sub-millisecond Julia EMT DAE control step window of $T_{\text{trans}} = 500$ μs, the network constant is derived exactly as:

$$a = \frac{5 \ \mu\text{s}}{500 \ \mu\text{s}} = 0.01 \tag{4}$$

The absolute throughput performance (S) as a function of offered traffic load (G) under this parameter is governed by the non-linear invariant:

$$S = \frac{G \cdot e^{-aG}}{G \cdot (1 + 2a) + e^{-aG} - (1 - e^{-aG})} \tag{5}$$

When $a = 0.01$, the system achieves optimal channel utilization efficiency, allowing our decentralized MARL agents to maximize collective rewards and execute randomized delay gating before control-loop interaction chaos collapses the grid.

5 Black-Hunter-Pan Multi-Ontology Dialogue Poset Strategy

When independent corporate data centers must resolve grid access boundaries without data leakage or brute-force database merges, the deconfliction process is modeled as a preference-ordered dialogue game following the algebraic principles of **Black, Hunter & Pan**.

Agents exchange lightweight symbolic arguments formulated as support-claim tuples $\mathcal{A}_k = \langle \Phi, \alpha \rangle$ across a Model Context Protocol (MCP) tool channel. Let $\mathcal{D}_t$ represent the sequential state-history matrix of the active dialogue at turn t. The turn-taking strategy function $\mathcal{S}_x(\mathcal{D}_t)$ acts as a deterministic, preference-ordered selection transducer over the Description Logic poset, mapped via the following strict dominance hierarchy:

$$\mathcal{S}_x(\mathcal{D}_t) = \text{Scounter}_x(\mathcal{D}_t) \succ \text{Sarg}_x(\mathcal{D}_t) \succ \text{Sconcede}_x(\mathcal{D}_t) \succ \langle x, \text{close}, \gamma \rangle \tag{6}$$

where:

$$\text{Scounter}_x(\mathcal{D}_t) = \{\langle x, \text{posit}, \langle \Phi, \phi \rangle \rangle \mid \langle \Phi, \phi \rangle \text{ is a novel Undercutter counter-argument}\} \tag{7}$$

$$\text{Sarg}_x(\mathcal{D}_t) = \{\langle x, \text{posit}, \langle \Phi, \gamma \rangle \rangle \mid \gamma = \text{Topic}(\mathcal{D}_t)\} \tag{8}$$

$$\text{Sconcede}_x(\mathcal{D}_t) = \{\langle x, \text{concede}, \phi \rangle \mid \phi \in \Delta_x \cup \Pi \text{ and } \phi \text{ is relevant to } \gamma\} \tag{9}$$

Transmission bandwidth overhead $\mathcal{C}$ scales as a compact function of decision forest graph depth d rather than the raw magnitude of the extensional grounding data:

$$\mathcal{C}(\mathcal{T}) \propto \mathcal{O}(d \cdot \Sigma_{\text{tokens}}) \ll \mathcal{C}(\mathcal{E}) \propto \mathcal{O}(|\mathcal{E}| \cdot \text{Dim}_{\text{features}}) \tag{10}$$

This enables the multi-agent perimeter to achieve absolute strategic consensus $\langle x, \text{close}, \gamma \rangle \cap \langle y, \text{close}, \gamma \rangle$ over thin network channels while keeping proprietary Level 1 telemetry perfectly firewalled.